v1.0

Validation Report

Bolted Joint Pre-Design Tool · Pressure Vessel & Piping — VDI 2230 · EN 1591-1 · ASME PCC-1

Tool Version	ClampForce v1.0
Standard Reference	VDI 2230 · EN 1591-1:2013 · ASME PCC-1
Scope	3 Scenarios — CHECK / VDI phi+Thermal / DESIGN+Sealing

This document covers the validation study of the ClampForce v1.0 bolted joint analysis tool. Three independent scenarios were carried out in detail with analytical calculation steps and compared against tool outputs. In all scenarios, the tool results were in full agreement with the hand-verified theoretical values.

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SCENARIO 1

CHECK Mode — Basic Flange Joint Validation

This scenario validates the basic CHECK mode of ClampForce. Input values are taken from a typical industrial application sourced from ASME PCC-1 and Shigley's Mechanical Engineering Design (10th Ed., Example 8-5). A fixed load introduction factor C is used; thermal effects and leakage check are disabled.

1.1 Input Parameters

Parameter	Value
Internal Pressure	0.5 MPa
Gasket Inner Diameter (d_inner)	100 mm
Gasket Outer Diameter (d_outer)	140 mm
Bolt Size	M16
Strength Class	8.8 → proof = 580 MPa, A_s = 157 mm2
Preload Factor (k)	0.75
Preload Loss Factor	0.10
Number of Bolts (n)	8
Load Introduction Factor (C)	0.25
SF — Separation	1.5
SF — Stress	1.2

1.2 Analytical Calculation Steps

Step	Expression	Result
1	d_eff = (100 + 140) / 2	120.0 mm
2	A_eff = (pi/4) * 120^2	11 310 mm2
3	F_p = 0.5 * 11 310	5 655 N
4	F_pre_raw = 0.75 * 580 * 157	68 265 N
5	F_pre = 68 265 * (1 - 0.10)	61 438 N
6	F_bolt = 61 438 + 0.25*(5 655/8)	61 615 N
7	sigma = 61 615 / 157	392.5 MPa
8	Limit = 580 / 1.2	483.3 MPa
9	Utilization = 392.5 / 483.3 * 100	81.2 %
10	Total preload = 8 * 61 438	491 508 N
11	Margin = 491 508 - 5 655	+485 853 N

1.3 Check Results

Check	Calculated	Limit	Status
[1] Stress Check	392.5 MPa	483.3 MPa (proof/SF)	✓ OK
[2] Separation Check	491.5 kN	5.66 kN (F_p)	✓ OK

OVERALL RESULT ✓ **ACCEPTABLE**

SCENARIO 2

VDI phi Active + Thermal — Steel / Aluminum Composite Flange

This scenario addresses a steel-aluminum composite flange joint based on VDI 2230 Part 1 (2015), Section 5.3. The VDI phi stiffness ratio calculation and thermal load analysis are activated simultaneously. The scenario is designed to demonstrate how thermal effects can critically compromise joint integrity.

2.1 Input Parameters

Parameter	Value
Internal Pressure	1.0 MPa
Gasket Inner Diameter (d_inner)	80 mm
Gasket Outer Diameter (d_outer)	120 mm
Bolt Size	M12
Strength Class	8.8 → proof = 580 MPa, A_s = 84.3 mm2
Preload Factor (k)	0.75
Preload Loss Factor	0.10
Number of Bolts (n)	6
VDI phi	ACTIVE
E_bolt (Steel)	210 000 MPa
E_joint (Aluminum)	70 000 MPa
L_eff	40 mm
Thermal Analysis	ACTIVE
DeltaT	80 °C
L_joint	40 mm
alpha_bolt (Steel)	12 × 10E-6 /K
alpha_joint (Aluminum)	23 × 10E-6 /K
SF — Separation	1.5
SF — Stress	1.2

2.2 Analytical Calculation Steps

Step	Expression	Result
1	d_eff = (80 + 120) / 2	100.0 mm
2	A_eff = (pi/4) * 100^2	7 854 mm2

3	t_joint = max(120-80, 1)	40 mm
4	k_b = 210 000 * 84.3 / 40	443 325 N/mm
5	k_j = 70 000 * 7 854 / 40	13 744 500 N/mm
6	phi = k_b / (k_b + k_j)	0.0312
7	F_p = 1.0 * 7 854	7 854 N
8	F_pre_raw = 0.75 * 580 * 84.3	36 670 N
9	F_pre = 36 670 * 0.90	33 003 N
10	dd = (23e-6*40 - 12e-6*40) * 80	0.03520 mm
11	dF_th = dd / (1/k_b + 1/k_j)	+15 120 N
12	F_pre_thermal = 33 003 + 15 120	48 123 N
13	F_bolt = 48 123 + 0.0312*(7 854/6)	48 164 N
14	sigma = 48 164 / 84.3	571.3 MPa
15	Limit = 580 / 1.2	483.3 MPa

2.3 Check Results

Check	Calculated	Limit	Status
[1] Stress Check	571.3 MPa	483.3 MPa (proof/SF)	✗ **FAIL**
[2] Separation Check	288.7 kN	7.85 kN (F_p)	✓ **OK**

OVERALL RESULT ✗ **NOT ACCEPTABLE**

Note: The NOT ACCEPTABLE outcome of this scenario is expected and intentional. The same joint without thermal analysis yields an ACCEPTABLE result (sigma=392.5 MPa < 483.3 MPa). This result demonstrates that the tool correctly captures thermally induced preload increase and proves the critical importance of thermal analysis.

SCENARIO 3

DESIGN Mode + Sealing — High-Pressure Flange Design

This scenario covers a high-pressure application based on the EN 1591-1:2013 flange joint design standard. In DESIGN mode, the tool automatically determines the minimum number of bolts; bolt spacing check and spiral wound gasket leakage check are run simultaneously. The minimum gasket contact stress is taken as 30 MPa, a typical value per EN 13555.

3.1 Input Parameters

Parameter	Value
Internal Pressure	2.5 MPa
Gasket Inner Diameter (d_inner)	150 mm
Gasket Outer Diameter (d_outer)	190 mm
BCD (Bolt Circle Diameter)	280 mm
Bolt Size	M20
Mukavemet Sinifi	10.9 → proof = 830 MPa, A_s = 245 mm2
Preload Factor (k)	0.75
Preload Loss Factor	0.08
Mod	DESIGN
Leakage Check	ACTIVE
Min. Gasket Contact Stress	30 MPa (Spiral wound — EN 13555)
Spacing Factor	3.0 × d
SF — Separation	1.5
SF — Stress	1.2

3.2 Analytical Calculation Steps

Step	Expression	Result
1	d_eff = (150 + 190) / 2	170.0 mm
2	A_eff = (pi/4) * 170^2	22 698 mm2
3	F_p = 2.5 * 22 698	56 745 N
4	F_pre_raw = 0.75 * 830 * 245	152 512 N
5	F_pre = 152 512 * (1 - 0.08)	140 311 N
6	n_min = ceil(1.5 * 56 745 / 140 311)	1 (raw)

7	n = max(4, n_min) sp = pi*280/4	219.9 mm
8	sp_min = 3.0 * 20	60.0 mm
9	219.9 >= 60.0 → Spacing OK, n = 4	PASS
10	A_gasket = (pi/4)*(190^2 - 150^2)	10 681 mm2
11	Total_pre = 4 * 140 311	561 244 N
12	p_gasket = (561 244 - 56 745) / 10 681	47.2 MPa
13	F_bolt = 140 311 + 0.25*(56 745/4)	143 858 N
14	sigma = 143 858 / 245	587.2 MPa
15	Limit = 830 / 1.2	691.7 MPa

3.3 Check Results

Check	Calculated	Limit	Status
[1] Stress Check	587.2 MPa	691.7 MPa (proof/SF)	✓ OK
[2] Separation Check	561.2 kN	56.7 kN (F_p)	✓ OK
[3] Leakage Check	47.2 MPa	30.0 MPa (EN 13555)	✓ OK
[4] Spacing Check	219.9 mm	60.0 mm (min.)	✓ OK

DESIGN MODE RESULTS	Min. n (raw): 1	Selected n: 4	Spacing: 219.9 mm

OVERALL RESULT ✓ **ACCEPTABLE**

GENERAL ASSESSMENT

Conclusion and Validation Summary

#	Scenario	Mod	Feature	Arac Resultu	Match
1	Basic Flange	CHECK	Fixed C, n=8	ACCEPTABLE	✓ FULL
2	Steel/Al. Flange	CHECK	VDI phi + Thermal	NOT ACCEPTABLE	✓ FULL
3	High-Pressure Flange	DESIGN	Sealing + Spacing	ACCEPTABLE, n=4	✓ FULL

In the three independent validation scenarios conducted, the ClampForce v1.0 analysis engine showed full agreement with the analytical reference calculations per VDI 2230 and EN 1591-1:2013 standards. The tool correctly computes pressure force, preload values, phi stiffness ratio, thermal load change, and gasket contact stress. In DESIGN mode, the minimum bolt count and spacing check produce the expected results.

The intentional NOT ACCEPTABLE result of Scenario 2 is a significant indicator that the tool correctly identifies critical safety conditions.

REFERENCES

[1]	**VDI 2230 Part 1:2015**	Systematische Berechnung hochbeanspruchter Schraubenverbindungen
[2]	**EN 1591-1:2013**	Flanges and their joints — Design rules for gasketed circular flange connections
[3]	**ASME PCC-1-2019**	Guidelines for Pressure Boundary Bolted Flange Joint Assembly
[4]	**Shigley's MED, 10th Ed.**	Budynas & Nisbett — Bolted joint analysis, Example 8-5
[5]	**EN 13555:2014**	Gasket parameters and test procedures relevant to the design rules for gasketed circular flange connections